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1. Design Explanation – Smart Water Gate System

Overview:

The system is a **vertically sliding water gate** used to regulate water flow in paddy fields. It uses a **NEMA 17 stepper motor** connected to a **lead screw**, which lifts or lowers the gate panel. The design ensures precision, automation, and protection against over-irrigation or water shortage.

Key Parts & Their Functions:

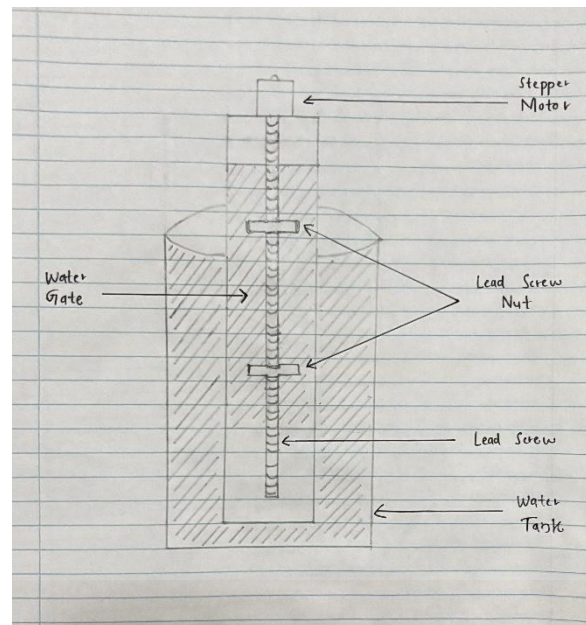


Figure 1: Water tank with water gate

• Water Tank / Canal

- **Purpose:** Holds or channels the irrigation water.
- **Description:** This is the section from which water is either held back or released depending on the gate position.
- **Placement:** Below and behind the gate door.

• Gate Panel (Sliding Door)

- **Purpose:** Acts as a barrier that controls water flow.
- **Material Suggestion:** Stainless steel, acrylic, or treated wood (to resist corrosion and pressure).
- **Movement:** Moves vertically up to open and down to close.

- **Lead Screw**

- **Purpose:** Converts the motor's rotational motion into linear motion to raise/lower the gate.
- **Type:** TR8x8 or T8 threaded rod.
- **Mounted Between:** Stepper motor (top) and gate nut (fixed to panel).

- **Lead Screw Nut**

- **Purpose:** Attached to the gate; travels along the screw when it rotates.
- **Note:** This nut moves up/down while preventing rotation. It's critical that it's fixed to the gate and constrained against rotation.

- **NEMA 17 Stepper Motor**

- **Purpose:** Provides precise rotational motion to the lead screw.
- **Control:** Driven by Raspberry Pi (through a driver like TMC2209 or A4988).
- **Placement:** Top of the gate frame.

Water Level Sensor

- **Purpose:** Measures the current water level in the tank or canal.
- **Use:** Helps determine whether the gate should be opened or closed based on tank water availability.
- **Type Suggestions:** Ultrasonic sensor (e.g., HC-SR04), float switch, or capacitive level sensor.
- **Placement:** Mounted inside or above the tank/canal facing the water surface.

Gate in Closed Position:

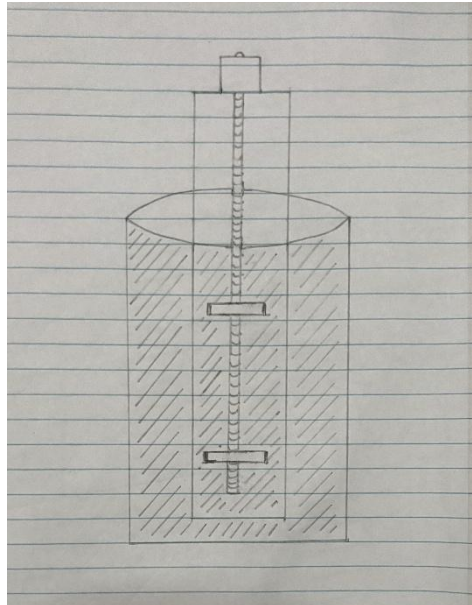


Figure 2: Gate in closed position

In the **closed state**, the gate is **fully lowered** into the water channel, **blocking the flow** of water completely.

Gate in Open Position:

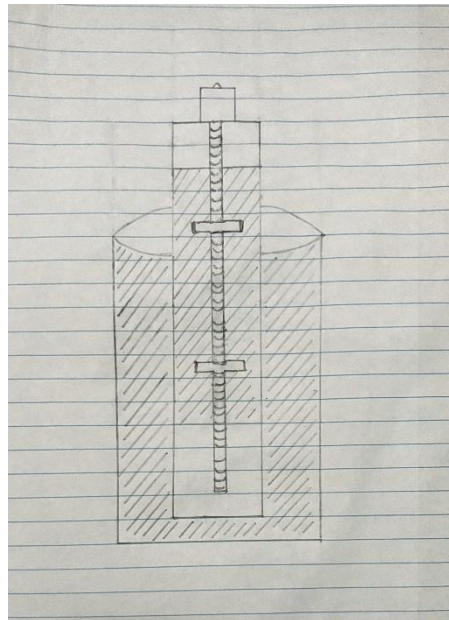


Figure 3: Gate in open position

In the **open state**, the gate is **lifted vertically** by the motor, allowing **uninterrupted water flow** through the channel.

2. Design Explanation - Pot-Based Paddy Irrigation Prototype

Objective:

This setup is used to simulate a real-world paddy field scenario in a controlled environment using a **plastic pot**. It helps test the functionality and responsiveness of the **smart water gate system** by monitoring moisture levels and rain detection.

Components & Their Functions

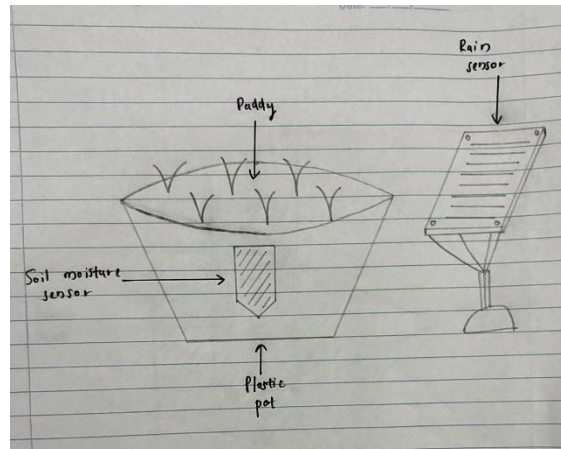


Figure 4: Pot based paddy irrigation

- **Plastic Pot**
 - **Purpose:** Serves as a miniature paddy field for testing.
 - **Contents:** Filled with soil and paddy plants.
 - **Advantage:** Portable, safe for sensor trials, and easy to monitor in indoor/outdoor conditions.
- **Paddy Plants**
 - **Purpose:** Represent the crop being cultivated.
 - **Significance:** Soil moisture and water level are monitored to ensure the ideal growing environment.
- **Soil Moisture Sensor**
 - **Placement:** Inserted into the soil inside the plastic pot.
 - **Purpose:** Measures real-time soil moisture content.
 - **Output:** Analog or digital values sent to the controller (Raspberry Pi or Arduino).
 - **Use Case:** Triggers irrigation when moisture falls below a defined threshold.
- **Rain Sensor (YL-83)**

- **Placement:** Positioned next to the pot in an elevated and exposed angle.
- **Purpose:** Detects the presence of rain.
- **Output:** Sends a digital LOW signal when raindrops are detected.
- **Use Case:** Prevents irrigation during rainfall to conserve water.

3. Integrated System Design: Water Gate to Pot-Based Irrigation

The following diagram represents the **complete integrated setup** combining the smart water gate system and the pot-based paddy irrigation unit. This configuration demonstrates a **closed-loop intelligent irrigation system** where decisions are made based on real-time sensor data and predictive modeling.

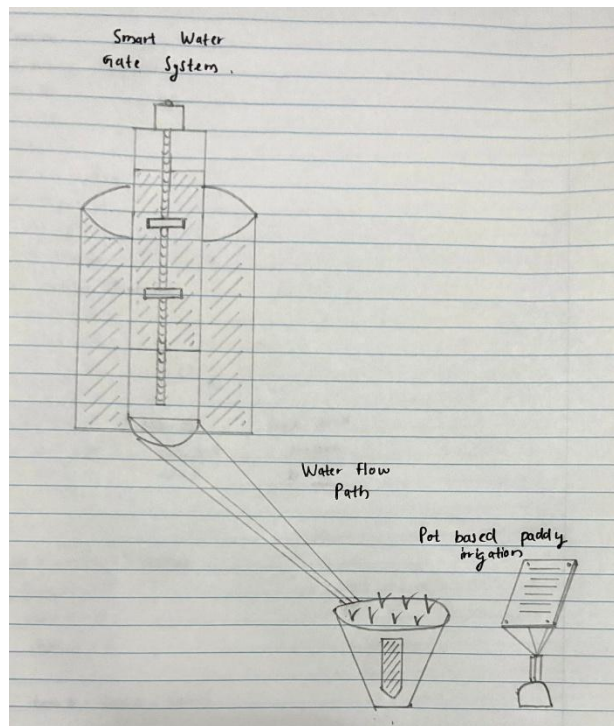


Figure 5: Water gate to pot based irrigation

Description of the Integrated System:

- On the **left side**, the **Smart Water Gate System** is installed over a cylindrical water tank. It consists of a NEMA 17 stepper motor and a lead screw mechanism that lifts or lowers the gate vertically to regulate water flow.
- Inside the tank, a **Water Level Sensor** is positioned to monitor the available water. This data is crucial for **predicting whether irrigation can proceed**.

- When the gate is opened, water exits the tank via the **Water Flow Path**—a sloped channel or pipe that delivers water to the crop area.
- On the **right side**, the **Pot-Based Paddy Irrigation Unit** is positioned to receive this water. It includes:
 - A **Soil Moisture Sensor** embedded in the soil to detect real-time moisture levels.
 - A **Rain Sensor** placed nearby to detect rainfall events.
 - A **plastic pot** filled with soil and planted with paddy to simulate field conditions.

Predictive Model for Water Requirement:

To enhance automation, a **machine learning model** is used to **predict the current water requirement** for the paddy crop. The model takes the following inputs:

- **Soil Moisture Level**
- **Water Level in the Tank**
- **Other (temperature, future rainfall etc.)**

Based on these inputs, the model outputs whether irrigation is currently needed. This decision is then used to control the gate mechanism.

System Behavior:

Condition	System Response
Model predicts water is needed and no rain detected	Gate opens, water flows to the pot
Soil is moist or rain is detected	Gate remains closed
The tank water level is too low	Gate remains closed, irrigation blocked

- The **gate closes automatically** once the moisture level is sufficient or if rainfall is detected.
- The system ensures **minimal water usage** by combining **sensor readings with model-based prediction**.

Key Integration Points:

Component	Connected To / Function
Smart Water Gate	Controls the physical release of water from the tank
Water Level Sensor	Sends real-time tank level data to the prediction model
Soil Moisture Sensor	Monitors current soil dryness and sends data to the model
Rain Sensor	Prevents water gate operation during natural rainfall
Water Flow Path	Transfers water from the tank directly to the paddy pot
Predictive Model	Computes whether irrigation is required (based on sensor input)
Paddy Pot	Final recipient of the irrigated water